IBA MINI PROJECT
SMART BUSINESS FAQ CHATBOT

Complete Manual: Excel + Python + Machine Learning + GitHub + Streamlit

Level: Beginner | Objective: Build and publish a small ML-powered business chatbot.

Important: No prior NLP knowledge is required. TF-IDF is used only as a simple way to convert customer questions into numbers.

1. What You Will Build

```
Customer Question
      ↓
Excel Business Data
      ↓
Python + Pandas
      ↓
TF-IDF
      ↓
Logistic Regression
      ↓
Predicted Business Intent
      ↓
Predefined Response
      ↓
Streamlit Web App
      ↓
GitHub Repository
```

The project demonstrates the complete IBA journey: Business Problem → Data → Analysis/ML → Application → Business Value.

2. Business Problem

A retail company receives repetitive customer-support questions about orders, returns, refunds, payments, delivery, working hours and products. The company wants a simple chatbot that can recognize the type of question and provide an appropriate response.

The chatbot does not generate new answers. It classifies the question into an intent and retrieves a predefined business response.

3. Learning Outcomes

- Read business data from Excel using Pandas.
- Understand intent and classification.
- Convert text into numerical features using TF-IDF.
- Train a Logistic Regression model.
- Build a chatbot function.
- Create a GitHub repository.
- Create a Streamlit frontend.
- Understand how an ML model becomes a simple business application.

4. Tools Required

- GitHub account
- Python 3.x
- VS Code or GitHub Codespaces
- Excel
- Streamlit
- Pandas
- openpyxl
- Scikit-learn

5. Final Project Files

```
iba-smart-chatbot/
├── app.py
├── chatbot_data.xlsx
├── requirements.txt
└── README.md
```

All four files should ultimately be in the GitHub repository.

6. STEP 1 – Create the Excel Dataset

Create an Excel workbook named chatbot_data.xlsx. Create a sheet named Data. Use exactly these three columns:

```
Question | Intent | Response
```

Enter this starter data:

Question	Intent	Response
Hello	greeting	Hello! Welcome to customer support. How can I help you?
Hi there	greeting	Hello! Welcome to customer support. How can I help you?
Good morning	greeting	Hello! Welcome to customer support. How can I help you?
Where is my order?	order_tracking	Please provide your order ID so that we can help you track your order.
How can I track my package?	order_tracking	Please provide your order ID so that we can help you track your order.
I want to track my order	order_tracking	Please provide your order ID so that we can help you track your order.
I want to cancel my order	cancel_order	You can request cancellation before the order is shipped. Please provide your order ID.
Can I cancel my purchase?	cancel_order	You can request cancellation before the order is shipped. Please provide your order ID.
Please cancel my order	cancel_order	You can request cancellation before the order is shipped. Please provide your order ID.
I want to return my product	return_product	You can request a return according to the return policy.

		Please provide your order details.
How can I return an item?	return_product	You can request a return according to the return policy. Please provide your order details.
Can I return this product?	return_product	You can request a return according to the return policy. Please provide your order details.
I want a refund	refund	Refunds are processed after approval. Please provide your order ID.
When will I get my refund?	refund	Refunds are processed after approval. Please provide your order ID.
I need my money back	refund	Refunds are processed after approval. Please provide your order ID.
Do you accept UPI?	payment_methods	We accept UPI, cards and other available online payment methods.
Can I pay by card?	payment_methods	We accept UPI, cards and other available online payment methods.
What payment methods do you accept?	payment_methods	We accept UPI, cards and other available online payment methods.
How long does delivery take?	delivery	Standard delivery usually takes 3 to 7 business days depending on location.
When will my order arrive?	delivery	Standard delivery usually takes 3 to 7 business days depending on location.
Do you offer home delivery?	delivery	Standard delivery usually takes 3 to 7 business days depending on location.
What are your working hours?	working_hours	Customer support is available from 9 AM to 6 PM, Monday to Saturday.
When are you open?	working_hours	Customer support is available from 9 AM to 6 PM, Monday to Saturday.
When can I contact support?	working_hours	Customer support is available from 9 AM to 6 PM, Monday to Saturday.
Tell me about this product	product_information	Please provide the product name or product ID for more information.
What are the product features?	product_information	Please provide the product name or product ID for more information.
I need product information	product_information	Please provide the product name or product ID for more information.

For the final student project, expand the dataset to at least 100 questions and at least 8 intents.

7. STEP 2 – Create GitHub Repository

1. Log in to GitHub.
2. Choose New repository.
3. Name it iba-smart-chatbot.
4. Choose Public if required by your instructor.
5. Create the repository.
6. Upload chatbot_data.xlsx.

8. STEP 3 – requirements.txt

Create a file named requirements.txt and paste:

```
pandas
openpyxl
scikit-learn
streamlit
```

9. STEP 4 – app.py: COMPLETE APPLICATION CODE

Create app.py and paste the following complete code:

```
import streamlit as st
import pandas as pd
import re
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

# Page settings
st.set_page_config(
    page_title="Smart Business FAQ Chatbot",
    page_icon="  "
)

st.title("  Smart Business FAQ Chatbot")
st.write(
    "Ask a question about orders, returns, refunds, "
    "payments, delivery or products."
)

# Read Excel
df = pd.read_excel("chatbot_data.xlsx")

# Remove incomplete records
df = df.dropna(subset=["Question", "Intent", "Response"])

# Basic text cleaning
def clean_text(text):
    text = str(text).lower()
    text = re.sub(r"[^a-zA-Z0-9\s]", "", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

df["Clean_Question"] = df["Question"].apply(clean_text)

# Features and target
X = df["Clean_Question"]
y = df["Intent"]
```

```
# Convert text to numerical features
vectorizer = TfidfVectorizer()
X_tfidf = vectorizer.fit_transform(X)

# Train Machine Learning model
model = LogisticRegression(max_iter=1000)
model.fit(X_tfidf, y)

# Create response dictionary from Excel
responses = (
    df.groupby("Intent")["Response"]
      .first()
      .to_dict()
)

# Chatbot function
def chatbot(question):
    question = clean_text(question)
    question_vector = vectorizer.transform([question])
    intent = model.predict(question_vector)[0]
    response = responses[intent]
    return intent, response

# User interface
question = st.text_input("Enter your question:")

if question:
    intent, response = chatbot(question)
    st.success(response)
    st.info("Predicted Intent: " + intent)

# Dataset information
with st.expander("View Dataset Information"):
    st.write("Training questions:", len(df))
    st.write("Number of intents:", df["Intent"].nunique())
    st.dataframe(df[["Question", "Intent"]])

st.caption("IBA Machine Learning Mini Project")
```

10. How app.py Works

- pd.read_excel() reads the business data from Excel.
- clean_text() performs basic cleaning.
- X contains questions and y contains intent labels.
- TfidfVectorizer converts text into numbers.
- LogisticRegression learns the relationship between question features and intents.
- chatbot() predicts an intent and returns the response.
- Streamlit creates the browser-based frontend.

11. STEP 5 – README.md

Create README.md and paste the following. Replace the details with your own.

```
# Smart Business FAQ Chatbot

## Business Problem
A retail company receives repetitive customer-support questions.
The chatbot classifies customer questions and provides predefined responses.
```

```
## Technologies
Python, Pandas, Scikit-learn, TF-IDF, Logistic Regression,
Excel, GitHub and Streamlit.

## Dataset
The application reads Question, Intent and Response from chatbot_data.xlsx.

## Machine Learning
TF-IDF converts text into numerical features.
Logistic Regression performs intent classification.

## How to Run

pip install -r requirements.txt

streamlit run app.py

## Author
Name:
Programme:
Semester:
```

12. STEP 6 – Run the Project Locally

If using VS Code/local Python, open the project folder in the terminal.

```
pip install -r requirements.txt
streamlit run app.py
```

A browser window will open with the Streamlit application. If it does not open automatically, use the local address displayed in the terminal.

13. STEP 7 – Test the Application

Try these questions:

- Where is my package?
- I need my money back.
- Can I pay using UPI?
- I want to return an item.
- When will my order arrive?
- What are your support hours?
- Tell me about the product.

Check both the chatbot answer and the predicted intent.

14. STEP 8 – Publish the Project on GitHub

7. Confirm that app.py is in the repository.
8. Confirm that chatbot_data.xlsx is in the repository.
9. Confirm that requirements.txt is present.
10. Confirm that README.md is present.
11. Commit all changes.
12. Open the repository and check that all four files are visible.

Your GitHub repository is now the source-code home of your project.

15. STEP 9 – Deploy with Streamlit

Use Streamlit Community Cloud or the deployment method specified by your instructor.

13. Sign in using GitHub.
14. Select your repository.
15. Select the main branch.
16. Select app.py as the application file.
17. Click Deploy.
18. Wait for the application to build.
19. Open and test the generated application link.

If deployment fails, first check that app.py and chatbot_data.xlsx are in the same repository level and that requirements.txt contains all required packages.

16. Critical File-Path Rule

The application contains:

```
pd.read_excel("chatbot_data.xlsx")
```

Therefore the Excel file must be available in the same repository/project folder as app.py. Do not use a path such as C:\Users\YourName\Desktop\chatbot_data.xlsx because that path will not exist on Streamlit Cloud.